Nirabhra Mandal, Ph.D.

Postdoctoral Researcher
Electrical and Computer Engineering, Purdue University
+1 (858) 257-8696 | mandaln@purdue.edu

Education

University of California, San Diego, USA 2021 – 2026
Doctor of Philosophy, Mechanical and Aerospace Engineering
GPA: 3.86/4

Indian Institute of Science, Bengaluru, India 2018 – 2021
Master of Technology (Research), Electrical Engineering
GPA: 8.6/10

Institute of Engineering & Management, Kolkata, India 2013 – 2017
Bachelor of Technology, Electrical Engineering
GPA: 9.28/10

Journal Articles

2026 Behaviorally Heterogeneous Multi-Agent Exploration Using Distributed Task Allocation, *IEEE Robotics and Automation Letters* *[url]*

2025 Distributed Task Allocation for Self-Interested Agents With Partially Unknown Rewards, *IEEE Transactions on Automatic Control* *[url]*

2024 Opinion Dynamics for Utility-Maximizing Agents: Exploring the Impact of the Resource Penalty, *IEEE Transactions on Control of Network Systems* *[url]*

2021 Dynamics of a Stratified Population of Optimum-Seeking Agents on a Network – Part II: Steady-State Analysis, *IEEE Transactions on Control of Network Systems* *[url]*

2021 Dynamics of a Stratified Population of Optimum-Seeking Agents on a Network – Part I: Modeling and Convergence Analysis, *IEEE Transactions on Control of Network Systems* *[url]*

Journal Articles Submitted

2026 Distributionally Robust Nash Equilibrium Seeking with Partial Observations and Distributed Communication, *arXiv preprint arXiv:2605.15534* *[url]*

Conference Publications

2025 Multi-Topic Projected Opinion Dynamics for Resource Allocation, *IEEE Conference on Decision and Control (CDC)* *[url]*

2023 Opinion Dynamics for Utility-Maximizing Agents With Heterogeneous Resources, *American Control Conference (ACC) [url]*

2023 Control of Discrete-Time LTI Systems Using Stochastic Ensemble Systems, *SIAM Conference on Control and Applications [url]*

2020 Evolution of a Population of Selfish Agents on a Network, *IFAC World Congress [url]*

2016 MATLAB/Simulink Model of a Stand-Alone Solar PV System With MPPT-Enabled Optimized Power Conditioning Unit, *IEEE 7th Annual IEMCON*

2016 Mathematical Modeling and Characteristic Analysis of a Solar PV Cell, *IEEE 7th Annual IEMCON*

Book Chapters

2020 A Novel Three-Phase 5-Level Inverter Control and Its Performance Analysis for a Grid-Connected Solar PV Power System, *Advances in Energy Research, Vol. 2, Springer [url]*

2020 A Novel ANN-SMC-Based Maximum Power Point Tracking for Efficient DC-Stage Conversion of a Solar PV Power Plant, *Advances in Energy Research, Vol. 2, Springer [url]*

Notable Awards and Achievements

2024 Hay Medal for the Best Thesis in M. Tech (Res), Indian Institute of Science (IISc), Bengaluru

2019 Fellowship Grant from the Centre for Networked Intelligence (a Cisco CSR Initiative), Indian Institute of Science, Bangalore

2017 Best Student Award, Institute of Engineering & Management (IEM), Kolkata, awarded by Tata Consultancy Services

2017 Dr. A. P. J. Abdul Kalam Award for Best Techno-Wiz, Institute of Engineering & Management (IEM), Kolkata (2016–2017)

Internship and Lab Experience

Army Research Laboratory, Adelphi, MD, USA 2024
Summer Intern
Implemented distributed mapping algorithm on ROS simulations. Tested performance of this algorithm on Python/ROS environments. Integrated the code with internal code base.

Army Research Laboratory, Adelphi, MD, USA 2023
Summer Intern
Worked with internal code base built on top of the ROS environment. Implemented distributed task allocation algorithm on ROS simulations. Tested the algorithm performance with standard task allocation algorithms.

Multi Agent Robotics (MURO) Lab, UCSD 2021 - 2026
Lab Lead
Mentored several students across various projects related to implementation of algorithms on robots. Jointly created the MURO-bridge repository, a ROS2 framework for simulating and implementing distributed algorithms for robotic deployment.

Teaching Activities

Robot Motion Planning, MAE 242 Spring 2026
Teaching Assistant
Assisted in creating lesson plans and taught several classes. Assisted in setting up homework assignments, solutions, and rubrics. Conducted office hours for the students and graded assignments and exams.

Robotic Planning & Estimation, MAE 145 Spring 2026
Teaching Assistant
Assisted in creating lesson plans and taught several classes. Hosted discussion hours to familiarize students with course prerequisites like Python coding, probability, and random variables. Assisted in setting up homework assignments, solutions, and rubrics. Conducted office hours for the students and graded assignments and exams.

Robot Motion Planning, MAE 242 Fall 2024
Teaching Assistant
Assisted in setting up homework assignments, solutions, and rubrics. Taught several classes and graded assignments and exams.

Mentorship

- Jonathan Speaks (UCSD MAE, MS, Fall 2025 – Present)
- Bernie Zang (UCSD MAE, MS, Fall 2025 – Present)
- Cheng-Chou Chang (National Sun Yat-sen University, BTech, Summer 2025 - Present)
- Guanxiang Jing (UCSD MAE, MS, Spring 2025 – Summer 2026)
- Zeru Li (UCSD MAE, MS, Spring 2025 – Summer 2026)
- Sudhanshu Sankar (UCSD MAE, MS, Fall 2023 – Fall 2024)